

Ancient Clifford-Product Mean-Curvature Flow: Exact Lifespans, Focal Cylinders, and Minimal Stability

Route-A source-local certificate HCS-C319

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Abstract

For every $p, q \geq 1$, we solve mean-curvature flow inside the unit sphere for the full parallel family $S^p(\cos \theta) \times S^q(\sin \theta)$. The affine coordinate $y = \sin^2 \theta$ gives both ancient branches, their exact finite lifespans, and their focal collapses. We identify both Type-I round-cylinder blow-ups and prove exact area dissipation. At the minimal leaf we compute the entire separated Jacobi spectrum below and at zero, obtaining index $p + q + 3$ and nullity $(p + 1)(q + 1)$.

1 Exact phase portrait and lifespan

Put $n = p + q$, and write

$$\Sigma_\theta = S^p(\cos \theta) \times S^q(\sin \theta) \subset S^{n+1}, \quad 0 < \theta < \frac{\pi}{2}.$$

For $x \in S^p$, $z \in S^q$, choose $\nu = (-\sin \theta x, \cos \theta z)$. Our scalar convention is $\partial_t F = H\nu$; this is the negative area gradient for the signs below.

Theorem 1 (Complete product-family flow). *The principal curvatures are $\tan \theta$ with multiplicity p and $-\cot \theta$ with multiplicity q . Hence*

$$H = p \tan \theta - q \cot \theta, \quad y' = 2(ny - q), \quad y = \sin^2 \theta.$$

The unique stationary leaf has $y_ = q/n$. Every other solution is ancient,*

$$y(t) = \frac{q}{n} + \left(y_0 - \frac{q}{n}\right) e^{2nt}, \tag{1}$$

and has one finite forward endpoint. If $y_0 < q/n$, then

$$T_- = \frac{1}{2n} \log \frac{q}{q - ny_0}, \quad y(t) \downarrow 0,$$

and the S^p factor survives as the focal submanifold. If $y_0 > q/n$, then

$$T_+ = \frac{1}{2n} \log \frac{p}{ny_0 - q}, \quad y(t) \uparrow 1,$$

and the S^q factor survives. In both cases $y(t) \rightarrow q/n$ as $t \rightarrow -\infty$.

Proof. Differentiating $F = (\cos \theta x, \sin \theta z)$ gives $\partial_\theta F = \nu$. The two shape eigenvalues follow by differentiating ν along each factor. Thus $\theta' = H$, and multiplication by $2 \sin \theta \cos \theta$ yields the affine equation for y . Solving it gives (1); imposing $y = 0$ or $y = 1$ gives the stated times. Their logarithm arguments exceed one on the corresponding branches. Backward convergence, uniqueness of the stationary leaf, and the focal labels now follow directly. $\square$

The faces $y = 0, 1$ are focal submanifolds, not regular product hypersurfaces. The cases $p = 0$ or $q = 0$ are excluded degenerate sphere/double-cover geometries. The backward endpoint is at infinite time, not an added slice.

2 Type-I focal-cylinder geometry

The exact curvature and area density (up to the constant sphere volumes) are

$$|A|^2 = p \frac{y}{1-y} + q \frac{1-y}{y}, \quad (2)$$

$$\mathcal{A}(y) = (1-y)^{p/2} y^{q/2}. \quad (3)$$

Theorem 2 (Two Type-I limits and strict dissipation). *Every nonstationary solution of Theorem 1 satisfies*

$$\frac{d}{dt} \log \mathcal{A} = -H^2 < 0.$$

At a left collapse, $y = 2q(T_- - t) + O((T_- - t)^2)$ and

$$(T_- - t)|A|^2 \longrightarrow \frac{1}{2}, \quad \text{blow-up cylinder } S^q(\sqrt{2q}) \times \mathbb{R}^p.$$

At a right collapse, $1 - y = 2p(T_+ - t) + O((T_+ - t)^2)$, the same residue is $1/2$, and the cylinder is $S^p(\sqrt{2p}) \times \mathbb{R}^q$.

Proof. Taking the logarithmic derivative of (3) and substituting y' gives

$$\left(\frac{q}{2y} - \frac{p}{2(1-y)} \right) 2(ny - q) = -\frac{(ny - q)^2}{y(1-y)} = -H^2.$$

Linearising the exact affine solution at either finite endpoint gives the two expansions. Equation (2) then gives the Type-I residues. Under parabolic rescaling by $(T - t)^{-1/2}$, the collapsing sphere has squared radius $2q$ on the left and $2p$ on the right; the noncollapsing factor becomes its Euclidean tangent space. This proves the ordered cylinder statements. $\square$

3 Jacobi index, nullity, and Route-A boundary

At the stationary leaf the radii are $\sqrt{p/n}$ and $\sqrt{q/n}$, $|A|^2 = n$, and the normal Jacobi operator is $J = \Delta + |A|^2 + n = \Delta + 2n$. We fix the second-variation convention $\mathcal{Q}(f, f) = -\int_{\Sigma} f J f$.

Theorem 3 (Complete minimal stability count). *For spherical-harmonic degrees $\ell, m \geq 0$, the eigenvalue of $-\Delta$ is*

$$\lambda_{\ell m} = \frac{n\ell(\ell + p - 1)}{p} + \frac{nm(m + q - 1)}{q}.$$

Exactly the sectors $(0, 0), (1, 0), (0, 1)$ lie below $2n$, and only $(1, 1)$ equals $2n$. Therefore, for the quadratic-form convention associated to J , the Morse index is $n + 3$ and the nullity is $(p + 1)(q + 1)$.

Proof. Separation of variables on the two round factors gives $\lambda_{\ell m}$. For one factor, $\ell(\ell + p - 1)/p$ equals 0 at $\ell = 0$, equals 1 at $\ell = 1$, and exceeds 2 for $\ell \geq 2$; the same holds for m, q . Thus the stated four sectors exhaust the values at or below $2n$. Their harmonic multiplicities are $1, p + 1, q + 1$, and $(p + 1)(q + 1)$, proving the counts. $\square$

Finite evidence. The content-addressed atlas records 100 dimension pairs, 600 branch rows, and 3,600 spectral cells: 25,858 scalar leaves. An implementation-independent checker performs 22,738 checks, including independent normalized-area and dissipation calculations; SymPy closes 1,204 identities. Isolated replay is byte exact and 39 repaired-hash/parser mutations are rejected. These grids are regressions; Theorems 1–3 are analytic for all declared parameters.

Collision boundary. C281 treats intrinsic homogeneous Ricci flow, while the present evolution is extrinsic spherical mean-curvature flow. C314 concerns planar curve shortening, whereas the present theorem closes a higher-dimensional isoparametric hypersurface family and its minimal Jacobi spectrum.

Route-A result and nonclaims. Under NO_BAD_EULER_OR_ROOT_NUMBER, the strict tuple is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT).$$

The model supplies no arithmetic-prime owner, primitive-orbit Euler ledger, Euler product, target functional equation, counting law, or target-zero match. The Jacobi operator is source-local and only a formal spectral hint; it is not a Hilbert–Pólya operator. Route A is rejected and Route B stays locked. No classification beyond the two-factor family, target local data, root number, or automorphy is claimed.

AI use. A generative language model assisted drafting and code scaffolding. The displayed proofs, independent recomputation, hostile tests, and deterministic artifacts define the audit record.

Source lineage

References

- [1] F. dos Reis and K. Tenenblat, “The mean curvature flow by parallel hypersurfaces,” *Proceedings of the American Mathematical Society* 146 (2018), 4867–4878. DOI: [10.1090/proc/14178](https://doi.org/10.1090/proc/14178).
- [2] H.-L. Liu and C.-L. Terng, “The mean curvature flow for isoparametric submanifolds,” *Duke Mathematical Journal* 147 (2009), 157–179. DOI: [10.1215/00127094-2009-009](https://doi.org/10.1215/00127094-2009-009).